Unit 9, Lesson 8: Isolating a Quantity of Interest in a Quadratic Equation

Benchmark

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.

Learning Targets

- ☐ I can rearrange the vertex form of a quadratic equation to isolate a variable of interest.
- ☐ I can rearrange the factored form of a quadratic equation to isolate a variable of interest.
- ☐ I can rearrange the standard form of a quadratic equation to isolate a variable of interest.

9.8.1 Warm-Up: Bell Work

1. Solve $2x^2 - 12x + 14 = 2x - 6$ by factoring.

9.8.2 Collaboration: Forms of Quadratic Equations

The three different forms of quadratic equations are helpful in providing information about key features of the parabola. It may be helpful at times to understand how a quadratic equation can be rewritten to feature a specific variable.

Forms of Quadratic Equations

Vertex form of a quadratic equation is represented as $y = a(x-h)^2 + k$, where (h, k) is the vertex.

Factored form of a quadratic equation is represented as $y = a(x-p)(x-q)$, where $(p, 0)$ and $(q, 0)$ are the x-intercepts.

Standard form of a quadratic equation is represented as $y = ax^2 + bx + c$, where $(0, c)$ is the y-intercept.

1. Work with your partner to complete the table.

Equation	Variables of Interest	Variables that Look Easy to Isolate
$y = a(x-h)^2 + k$	a, h, k	
$y = a(x-p)(x-q)$	a, p, q	
$y = ax^2 + bx + c$	a, b, c	

2. Each partner should ask two different classmates which variables they selected. Record their variables and write your summary of their reasons. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Variables That Look Easy to Isolate	My Summary of their Reason	Initials

3. The variable a is the only variable that is common to all three forms. Which form appears to be the easiest to use to isolate the variable a?

9.8.3 Collaboration: Isolating the Variable a

1. Work with your partner to complete the steps for isolating the variable a.

Vertex Form	$y = a(x-h)^2 + k$
Subtract k from both sides.	$y - k = a(x-h)^2 + k - k$ $y - k = a(x-h)^2$
Divide both sides by $(x-h)^2$.	$\underline{\qquad\qquad} = \dfrac{a(x-h)^2}{(x-h)^2}$ $a = \underline{\qquad\qquad}$

Factored Form	$y = a(x-p)(x-q)$
Divide both sides by $(x-p)(x-q)$.	$\underline{\qquad\qquad\qquad} = \dfrac{a(x-p)(x-q)}{(x-p)(x-q)}$ $a = \underline{\qquad\qquad\qquad}$

Standard Form	$y = ax^2 + bx + c$
Subtract $bx + c$ from both sides.	$y - (\underline{\qquad\qquad}) = ax^2 + bx + c - (\underline{\qquad\qquad})$ $y - \underline{\qquad} - \underline{\qquad} = \underline{\qquad\qquad\qquad}$
Divide both sides by x^2.	$\dfrac{y - \quad - \quad}{\underline{\qquad\qquad\qquad}} = \underline{\qquad\qquad\qquad}$ $a = \dfrac{\quad - \quad - \quad}{\underline{\qquad\qquad\qquad}}$

2. Which form, if any, was the easiest to use to isolate the variable a?

9.8.4 Guided Instruction: Isolating the Variable x

1. Complete the table for isolating the variable x in vertex form.

Vertex Form	$\boldsymbol{y = a(x-h)^2 + k}$
Subtract k from both sides.	
Divide both sides by a.	
Take the ____________ of both sides. When taking the _______________ of both sides include $\pm$ to represent both of the possible outputs.	
Add h to both sides.	

The $\pm$ represents both possible results of a square root. Remember, when asked to find $\sqrt{36}$ it is understood that the principal square root, 6, is the answer, but that both 6 and -6 are correct because both values can be multiplied by itself to get 36. This can be written as $\sqrt{36} = \pm 6$.

2. Rewrite $x = -w \pm \sqrt{9y}$ as two equations.

3. Rewrite $x = h \pm \sqrt{\frac{y-k}{a}}$ as two equations.

4. Rewrite $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$ as two equations.

5. Complete the table for isolating the variable x in standard form.

Standard Form	$\boldsymbol{y = ax^2 + bx + c}$
From standard form the vertex of the parabola can be found using $\left(\frac{-b}{2a}, \frac{c}{a} - \frac{b^2}{4a^2}\right)$. Rewrite standard form to vertex form using this vertex.	$y = a(x-h)^2 + k$ $y = \left(x + \frac{b}{2a}\right)^2 + \frac{c}{a} - \frac{b^2}{4a^2}$
Subtract $\frac{c}{a}$ from both sides.	
Add the opposite of $-\frac{b^2}{4a^2}$ to both sides.	
Take the square root of both sides. When taking the square root of both sides, include $\pm$ to represent both of the possible outputs.	
Subtract $\frac{b}{2a}$ from both sides.	

9.8.5 Wrap-Up: Ticket Out the Door

1. Solve $y = 2(x-5)^2 + k$ for k.

2. Solve $y = -x^2 + bx + 7$ for b.